Check the Model

AP Statistics · Topic 1.10 (CED 2.11) · B: 2026-11-09 · E: 2026-12-02 · TEACHER KEY

CED TETHER

- ENDURING UNDERSTANDING: VAR-2 The normal distribution can be used to represent some population distributions.
- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- VAR-2.A Compare a data distribution to the normal distribution model. [Skill 2.D]
- VAR-2.A.1 A parameter is a numerical summary of a population. VAR-2.A.2 Some sets of data may be described as approximately normally distributed. A normal curve is mound-shaped and symmetric. The parameters of a normal distribution are the population mean, μ, and the population standard deviation, σ. VAR-2.A.3 For a normal distribution, approximately 68% of the observations are within 1 standard deviation of the mean, approximately 95% of observations are within 2 standard deviations of the mean, and approximately 99.7% of observations are within 3 standard deviations of the mean. This is called the empirical rule. VAR-2.A.4 Many variables can be modeled by a normal distribution.
- VAR-2.B Determine proportions and percentiles from a normal distribution. [Skill 3.A]
- VAR-2.B.1 A standardized score for a particular data value is calculated as (data value − mean)/(standard deviation), and measures the number of standard deviations a data value falls above or below the mean. VAR-2.B.2 One example of a standardized score is a z-score, calculated as z = (x - μ)/σ. A z-score measures how many standard deviations a data value is from the mean. VAR-2.B.3 Technology, such as a calculator, a standard normal table, or computer-generated output, can be used to find the proportion of data values located on a given interval of a normally distributed random variable. VAR-2.B.4 Given the area of a region under the graph of the normal distribution curve, it is possible to use technology to estimate parameters for some populations.
- VAR-2.C Compare measures of relative position in data sets. [Skill 2.D]
- VAR-2.C.1 Percentiles and z-scores may be used to compare relative positions of points within a data set or between data sets.

Essential question: When does a correct normal calculation justify a conclusion about actual data?

DOK-3 skill: 4.B · **FRQ pattern:** `is-the-normal-model-appropriate-adjudicate-model-vs-data`

DOK rationale: Part (c) separates model arithmetic from a warranted conclusion, tests the model against shape and an observed tail, and explains the consequence of a mismatch.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Read Ben's and Ava's lines; do not say whether the model fits.
- Video follow-along from minute 5 (normal curves, empirical rule, and z-scores).
- Board slide back up at minute 33. Circulate; require every model judgment to point to the histogram.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to using or rejecting the model before the video.

- Complete the video follow-along and check both model calculations.
- Finish (a)–(c); complete the exit line; turn in.

Questions to ask

- Which claim is about a normal curve, and which claim is about the 60 actual students?
- How many bars continue to the right after the peak? What would symmetry require on the left?
- How many students does 0.62% of 60 represent? How many do you actually see above 70?
- What must you inspect before typing values into normalcdf?

Adult role

Solo teacher. Circulate 5 pairs. Accept the arithmetic before pressing on the assumption; do not let 'correct z-score' stand in for a model check.

[WARN] WATCH FOR

- Treating the school's assertion of normality as evidence stronger than the histogram.
- Counting the 60–70 bar as above 70.
- Rejecting the model with no named feature or tail comparison.

SCORING PART (c) — E / P / I

Expected elements

- **separates-calculation-from-conclusion** (required) — Says the calculations are correct under the model but not warranted for the displayed data
- **cites-skew-or-shape** (required) — Cites the peak with a long right tail or lack of mound-shaped symmetry
- **compares-observed-tail** (required) — Compares 4 of 60 (6.7 percent) above 70 with the model's 0.62 percent
- **explains-model-check** (required) — Names the underestimated upper tail and says to check shape before normal methods

E	Separates correct model arithmetic from the data conclusion, cites shape and the 4-of-60 versus 0.62 percent tail discrepancy, and explains the underestimated tail and needed model check.
P	Rejects the model with only one feature, OR gives both features without explaining how correct arithmetic misleads, OR discusses model checking without comparing the tails.
I	Accepts the model because the calculations are correct, labels the histogram approximately normal, or gives no histogram evidence.

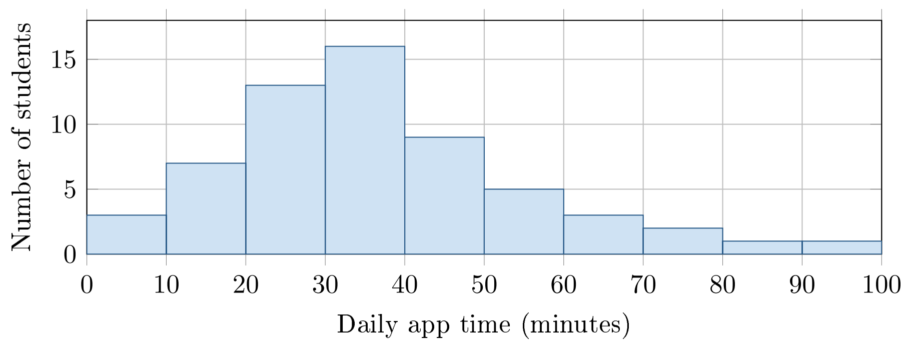
Common mistakes

- Treating a correct z-score as proof that the data are normal
- Saying "not normal" without naming skew, asymmetry, or the long tail
- Counting the 60–70 bin as above 70 or reporting 4 of 60 as 4%

[STAR] TODAY'S PROBLEM — annotated

A school says daily app time is approximately normal with mean 40 minutes and standard deviation 12 minutes. The histogram shows 60 students; no recorded time was exactly 70 minutes.

Ben: "About 16% should be above 52 minutes." **Ava:** "For 70 minutes, $z = 2.5$, so less than 1% should be above 70."



FIRST TAKE (5 min)

Does the normal model fit? Point to one histogram feature.
Key: Not graded. Expect trust in the stated model; preserve that tension.

Rules callout "USING A NORMAL MODEL (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** State μ and σ for the proposed model. What two shape features should approximately normal data have?

Key: $\mu = 40$ min and $\sigma = 12$ min. Approximately normal data should be mound-shaped and roughly symmetric.

DOK 2 **(b)** Check Ben's and Ava's model calculations. Then find the observed fraction of these 60 students above 70 minutes.

Key: 52 is 1σ above 40, so Ben's normal-model result is about 16%. For Ava, $P(Z > 2.5) \approx 0.0062 = 0.62\%$. The histogram has $2 + 1 + 1 = 4$ above 70, or $4/60 = 6.7\%$.

DOK 3 **(c)** Are the calculations warranted as conclusions about these students? Defend your decision with the histogram's shape and tail count. Explain how the relationship between a model and its data supports your decision, what a reader would conclude about long app times, and what must be checked before using normal methods.

Key: The arithmetic is correct for the model but not warranted for these data. The histogram peaks at 30–40 and has a long right tail; 4 of 60 (6.7%) exceed 70 versus the model's 0.62%. A calculation inherits its model assumption, so this model underestimates long app times. Check for a mound shape, symmetry, and contradictory skew, gaps, clusters, or outliers before using normal methods.

EXIT

One thing the video changed about my first take: ____________________